

Transmission Coefficient Through a Rectangular Potential Barrier

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Derivation of the Transmission Coefficient from Boundary Conditions

Consider a rectangular potential barrier

$$V(x) = \begin{cases} 0, & x < 0, \\ U, & 0 \leq x \leq L, \\ 0, & x > L, \end{cases}$$

with energy $E < U$. In the three regions the stationary state wavefunction is written as

$$\psi_I(x) = Ae^{ikx} + Be^{-ikx}, \quad x < 0, \quad (1)$$

$$\psi_{II}(x) = Ce^{iqx} + De^{-iqx}, \quad 0 \leq x \leq L, \quad (2)$$

$$\psi_{III}(x) = Fe^{ikx}, \quad x > L, \quad (3)$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad q = \sqrt{\frac{2m(E - U)}{\hbar^2}}.$$

For $E < U$ we write

$$q = i\kappa, \quad \kappa = \sqrt{\frac{2m(U - E)}{\hbar^2}},$$

and choose the normalization $A = 1$.

Step 1. Boundary Conditions at $x = 0$

Continuity of ψ and ψ' at $x = 0$ gives

$$\psi_I(0) = \psi_{II}(0) : \quad 1 + B = C + D, \quad (1)$$

$$\psi'_I(0) = \psi'_{II}(0) : \quad ik(1 - B) = iq(C - D). \quad (2)$$

Solving (1) and (2) for C and D yields:

$$C = \frac{1}{2} \left[(1 + B) + \frac{k}{q} (1 - B) \right], \quad (3)$$

$$D = \frac{1}{2} \left[(1 + B) - \frac{k}{q} (1 - B) \right]. \quad (4)$$

Step 2. Boundary Conditions at $x = L$

At $x = L$ we impose:

$$\psi_{II}(L) = \psi_{III}(L) : \quad C e^{iqL} + D e^{-iqL} = F e^{ikL}, \quad (5)$$

$$\psi'_{II}(L) = \psi'_{III}(L) : \quad iq \left[C e^{iqL} - D e^{-iqL} \right] = ik F e^{ikL}. \quad (6)$$

Eliminating C and D (by substituting (3) and (4) into (5), for example) leads to the transmission amplitude:

$$F = \frac{2kq e^{-ikL}}{2kq \cos(qL) - i(k^2 + q^2) \sin(qL)}. \quad (7)$$

Step 3. Substitution $q = i\kappa$

For $E < U$ substitute $q = i\kappa$. Then

$$\cos(qL) = \cosh(\kappa L), \quad \sin(qL) = i \sinh(\kappa L), \quad k^2 + q^2 = k^2 - \kappa^2.$$

Inserting these into (7) we have

$$\begin{aligned} F &= \frac{2k(i\kappa) e^{-ikL}}{2k(i\kappa) \cosh(\kappa L) - i(k^2 - \kappa^2)(i \sinh(\kappa L))} \\ &= \frac{2ik\kappa e^{-ikL}}{2ik\kappa \cosh(\kappa L) + (k^2 - \kappa^2) \sinh(\kappa L)}. \end{aligned} \quad (8)$$

Step 4. Obtaining the Transmission Coefficient

The transmission coefficient is defined as

$$T = |F|^2.$$

Before substituting $q = i\kappa$, expression (7) gives

$$T = \frac{4k^2 q^2}{\left[2kq \cos(qL) \right]^2 + \left[(k^2 + q^2) \sin(qL) \right]^2}.$$

Now substitute $q = i\kappa$ together with

$$\cos(qL) = \cosh(\kappa L), \quad \sin(qL) = i \sinh(\kappa L),$$

to obtain

$$\begin{aligned} T &= \frac{4k^2(i\kappa)^2}{\left[2k(i\kappa) \cosh(\kappa L)\right]^2 + \left[(k^2 - \kappa^2)(i \sinh(\kappa L))\right]^2} \\ &= \frac{4k^2(-\kappa^2)}{[2ik\kappa \cosh(\kappa L)]^2 + [i(k^2 - \kappa^2) \sinh(\kappa L)]^2} \\ &= \frac{4k^2\kappa^2}{4k^2\kappa^2 \cosh^2(\kappa L) + (k^2 - \kappa^2)^2 \sinh^2(\kappa L)}. \end{aligned} \tag{9}$$

Using the identity

$$(k^2 - \kappa^2)^2 \sinh^2(\kappa L) + 4k^2\kappa^2 \cosh^2(\kappa L) = 4k^2\kappa^2 + (k^2 + \kappa^2)^2 \sinh^2(\kappa L),$$

we rewrite (9) as

$$T = \frac{4k^2\kappa^2}{4k^2\kappa^2 + (k^2 + \kappa^2)^2 \sinh^2(\kappa L)}. \tag{10}$$

Step 5. Final Expression in Terms of E and U

Since

$$\begin{aligned} k^2 &= \frac{2mE}{\hbar^2}, \\ \kappa^2 &= \frac{2m(U - E)}{\hbar^2}, \\ k^2 + \kappa^2 &= \frac{2mU}{\hbar^2}, \end{aligned}$$

we have

$$4k^2\kappa^2 = \frac{16m^2 E(U - E)}{\hbar^4}, \quad (k^2 + \kappa^2)^2 = \frac{4m^2 U^2}{\hbar^4}.$$

Thus, (10) becomes

$$\begin{aligned} T &= \frac{\frac{16m^2 E(U - E)}{\hbar^4}}{\frac{16m^2 E(U - E)}{\hbar^4} + \frac{4m^2 U^2}{\hbar^4} \sinh^2(\kappa L)} \\ &= \left\{ 1 + \frac{U^2}{4E(U - E)} \sinh^2(\kappa L) \right\}^{-1}. \end{aligned} \tag{11}$$